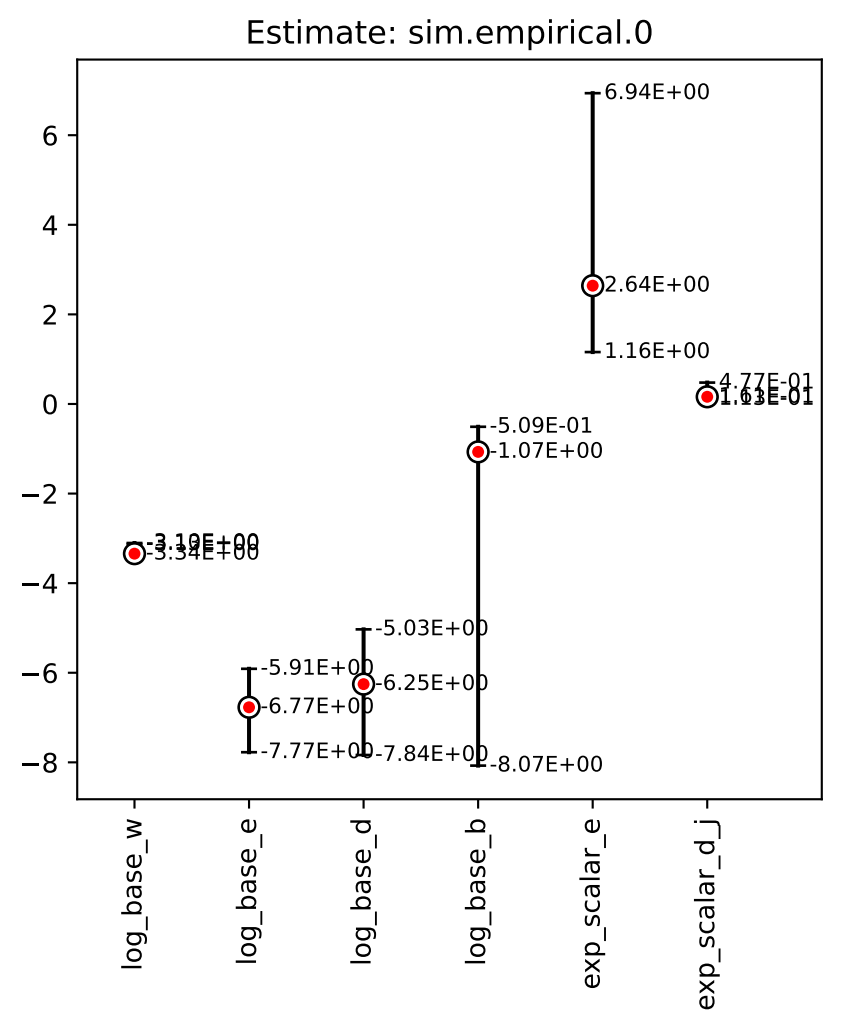
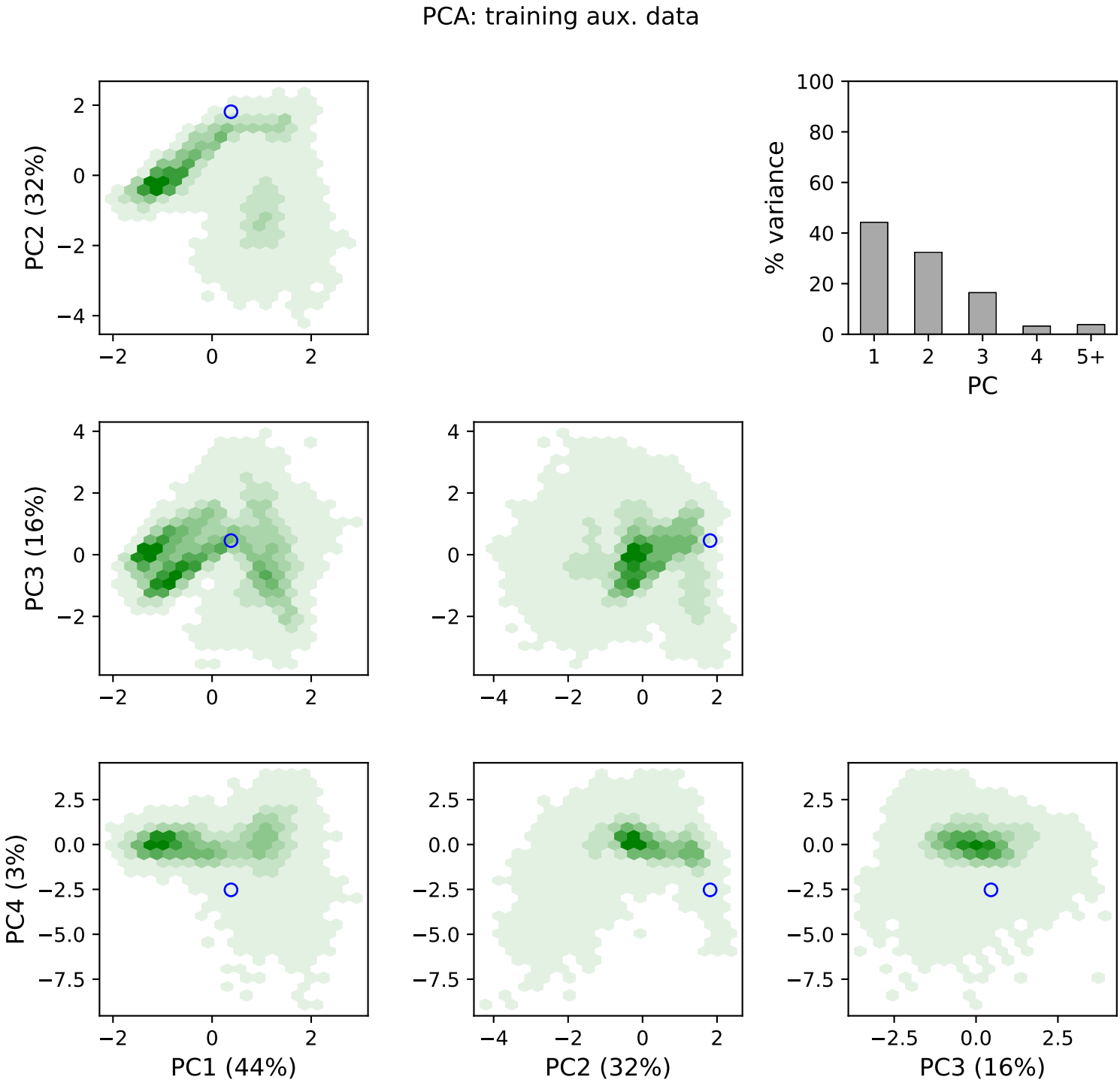
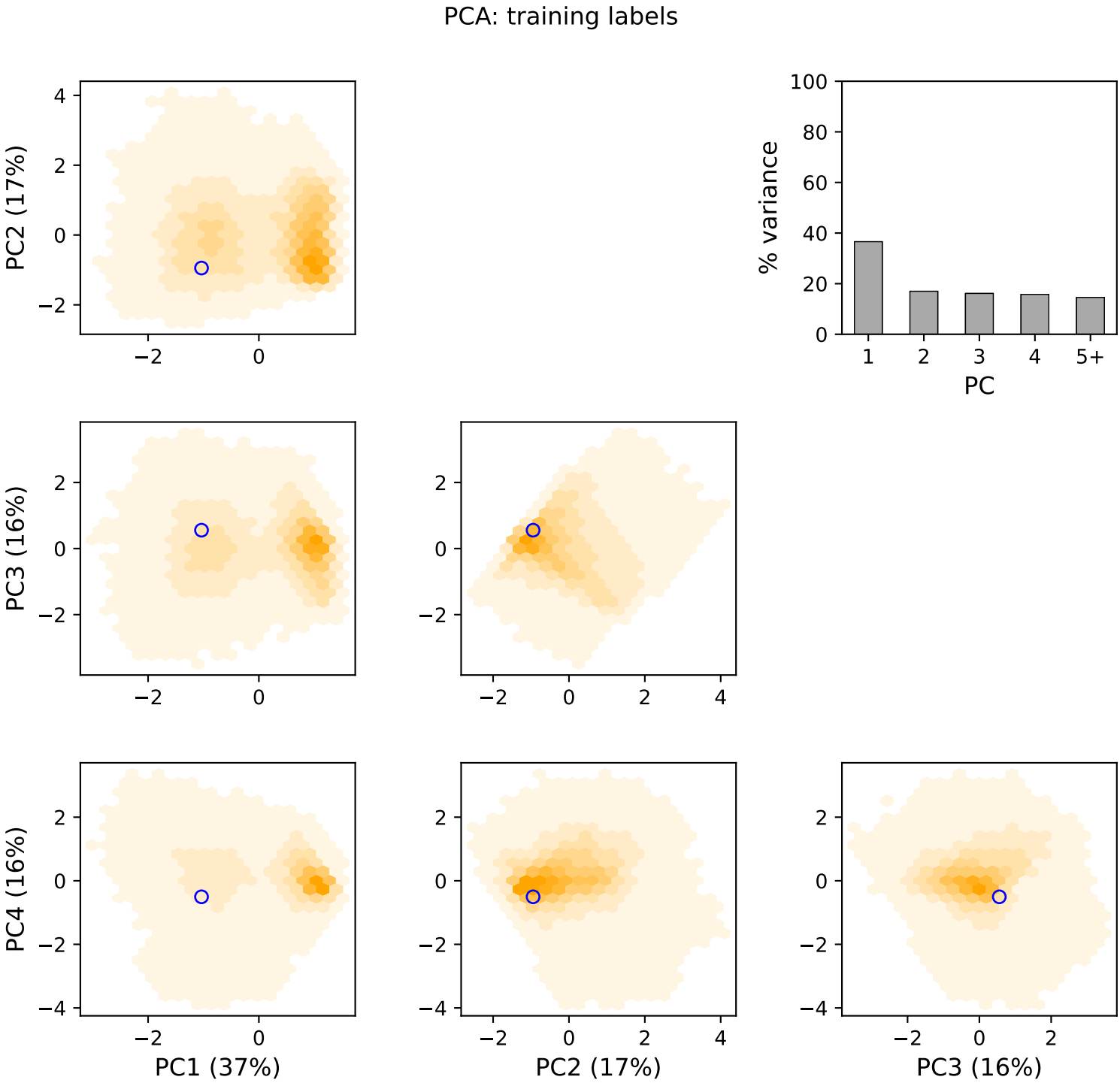
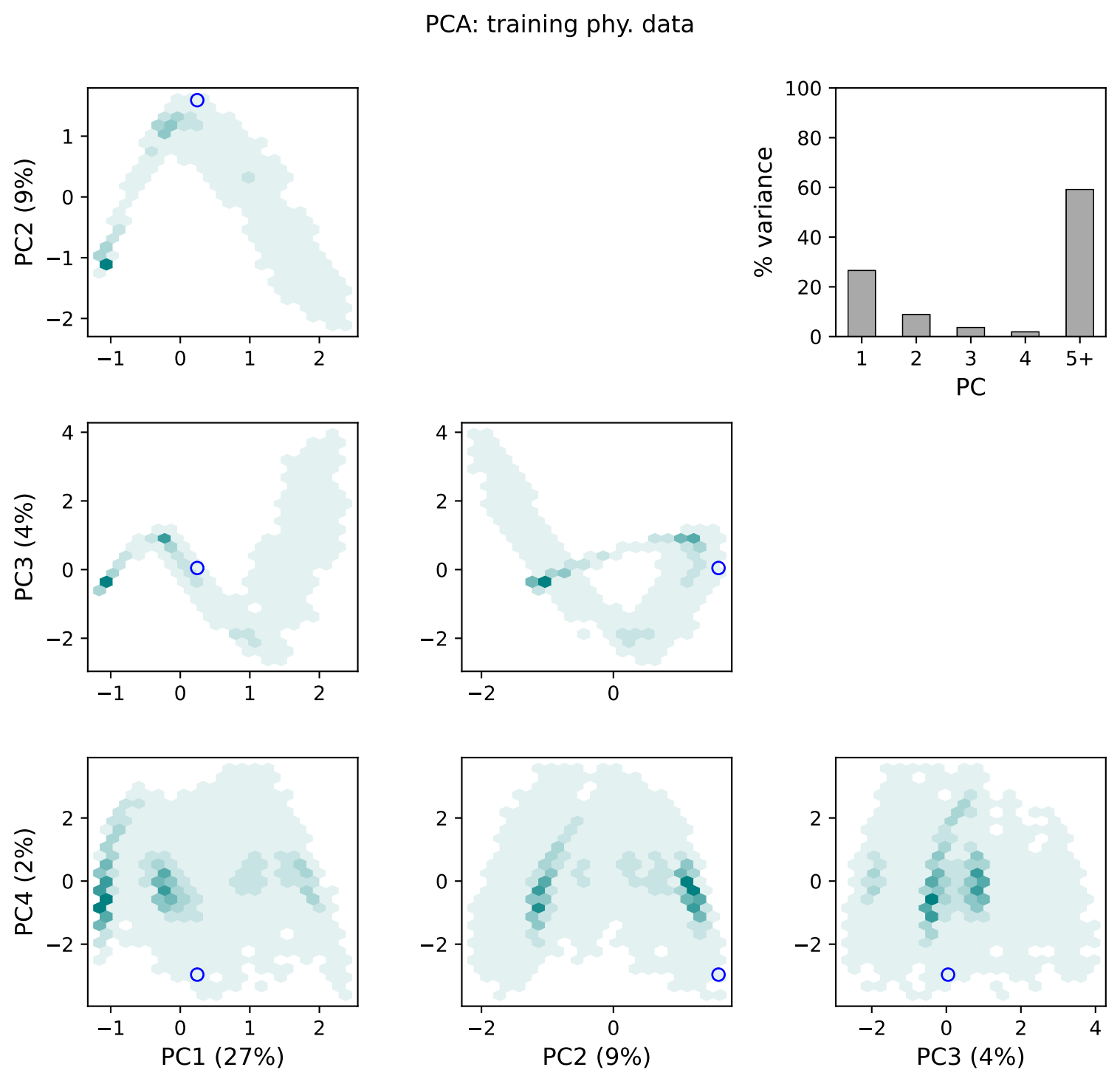
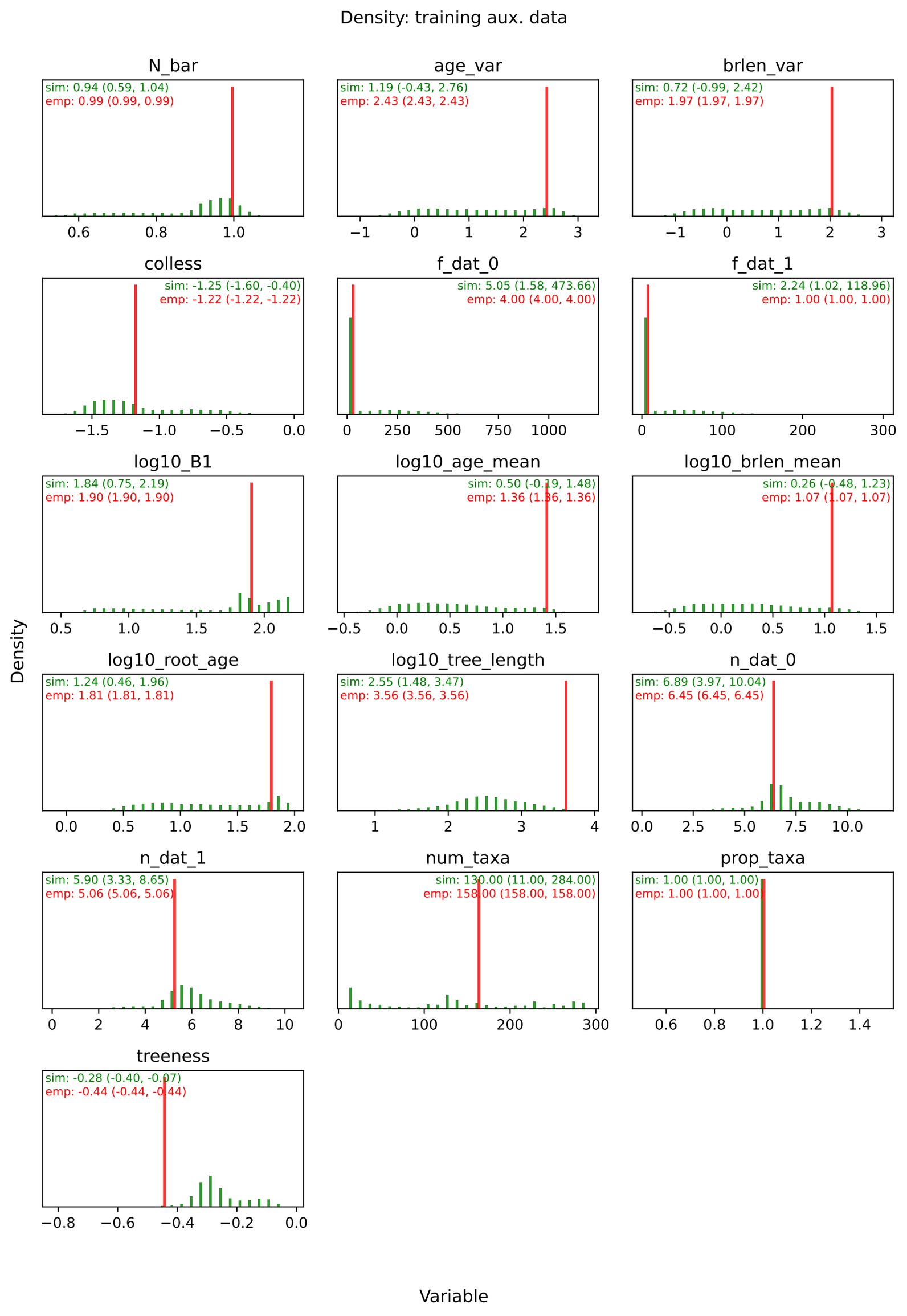
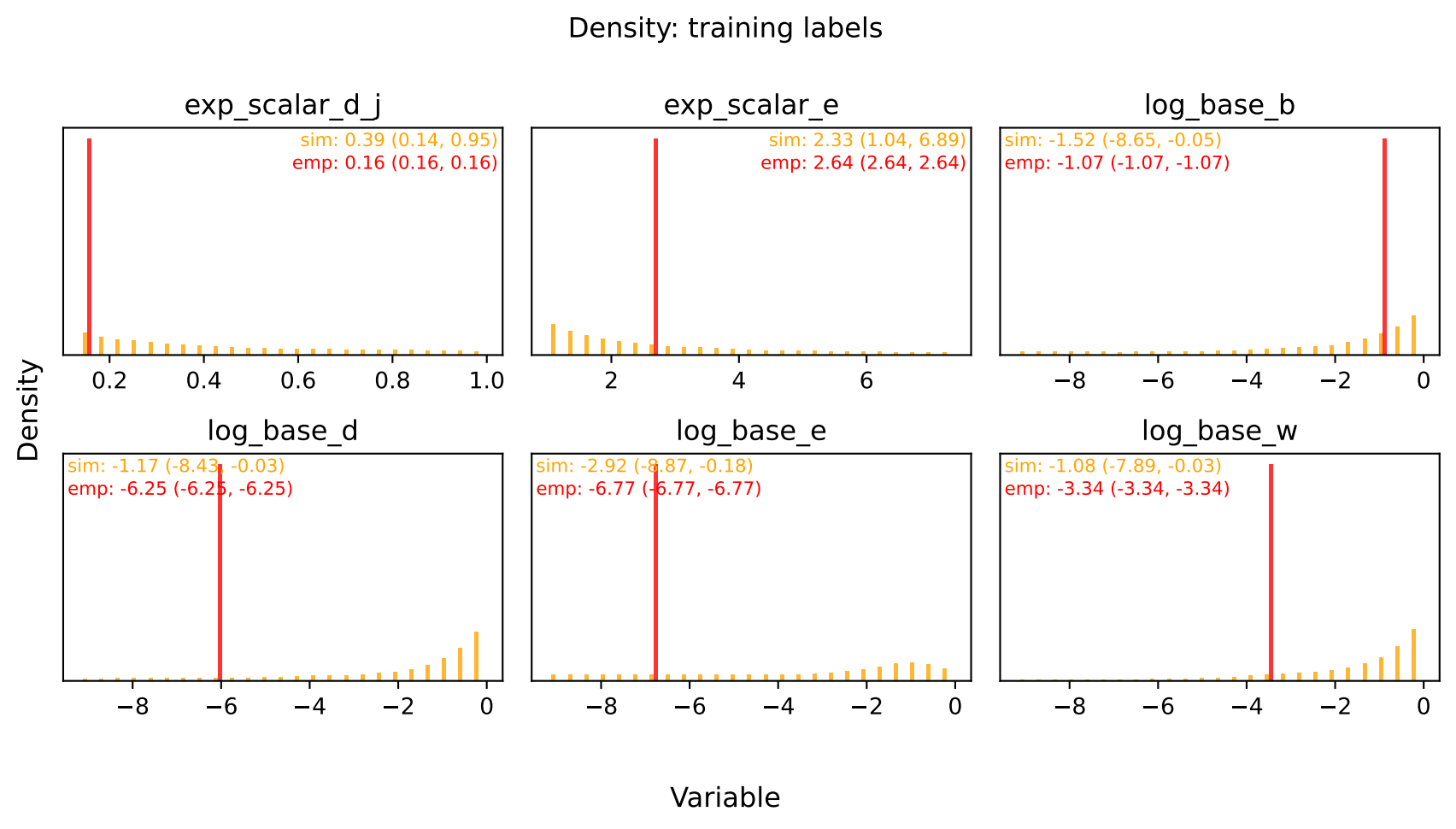

Estimate: sim.empirical.0
6.94E+00
2.64E+00
1.16E+00
-4.77E-01
-5.09E-01
-1.07E+00
-5.03E+00
-5.91E+00
-6.25E+00
-6.77E+00
-7.77E+00
-7.84E+00
-8.07E+00
6
4
2
0
−2
−4
−6
−8
log_base_w
log_base_e
log_base_d
log_base_b
exp_scalar_e
exp_scalar_d_j

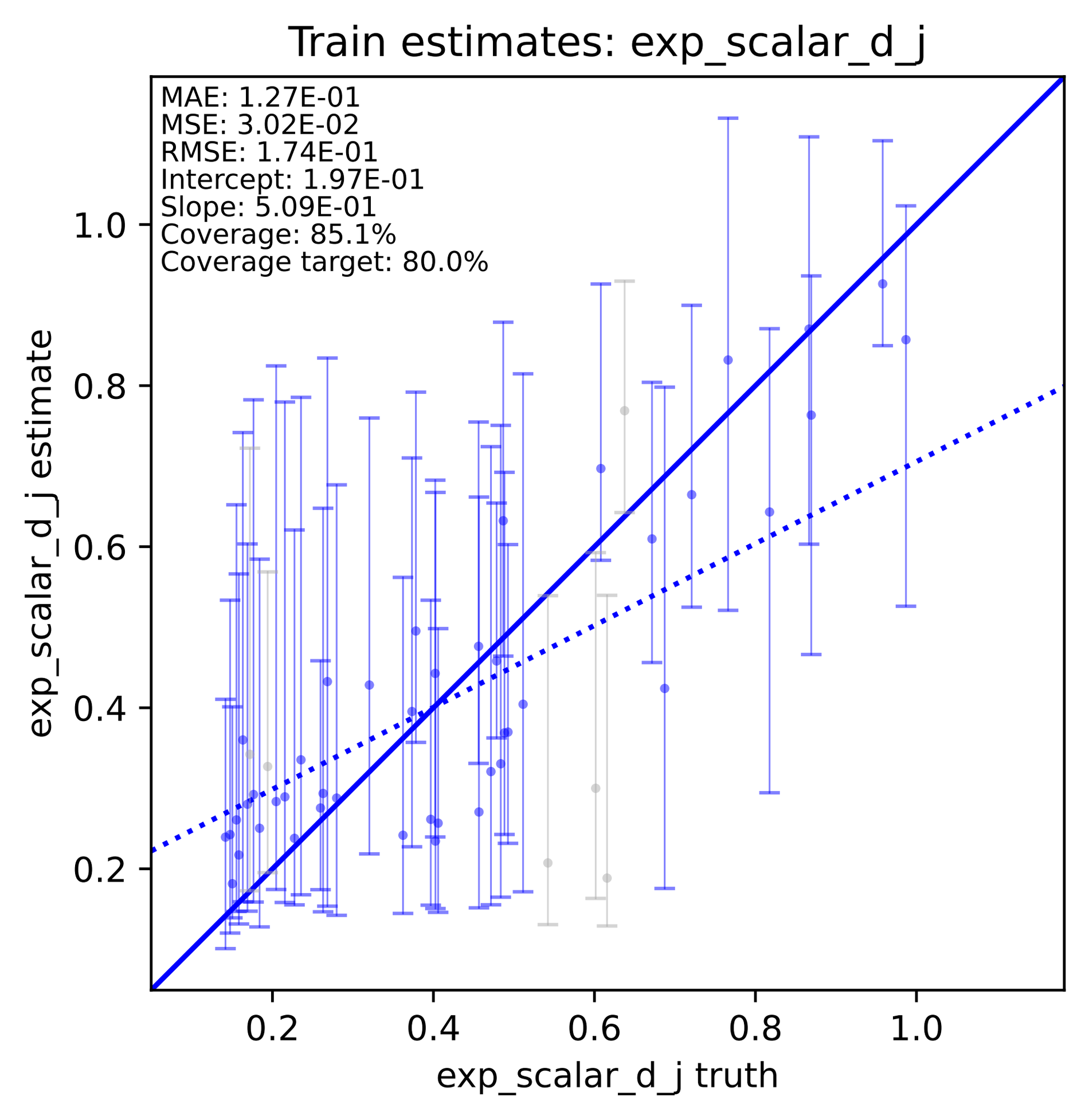

PCA: training aux. data
PC2 (32%)
% variance
PC
PC3 (16%)
PC4 (3%)
PC1 (44%)
PC2 (32%)
PC3 (16%)

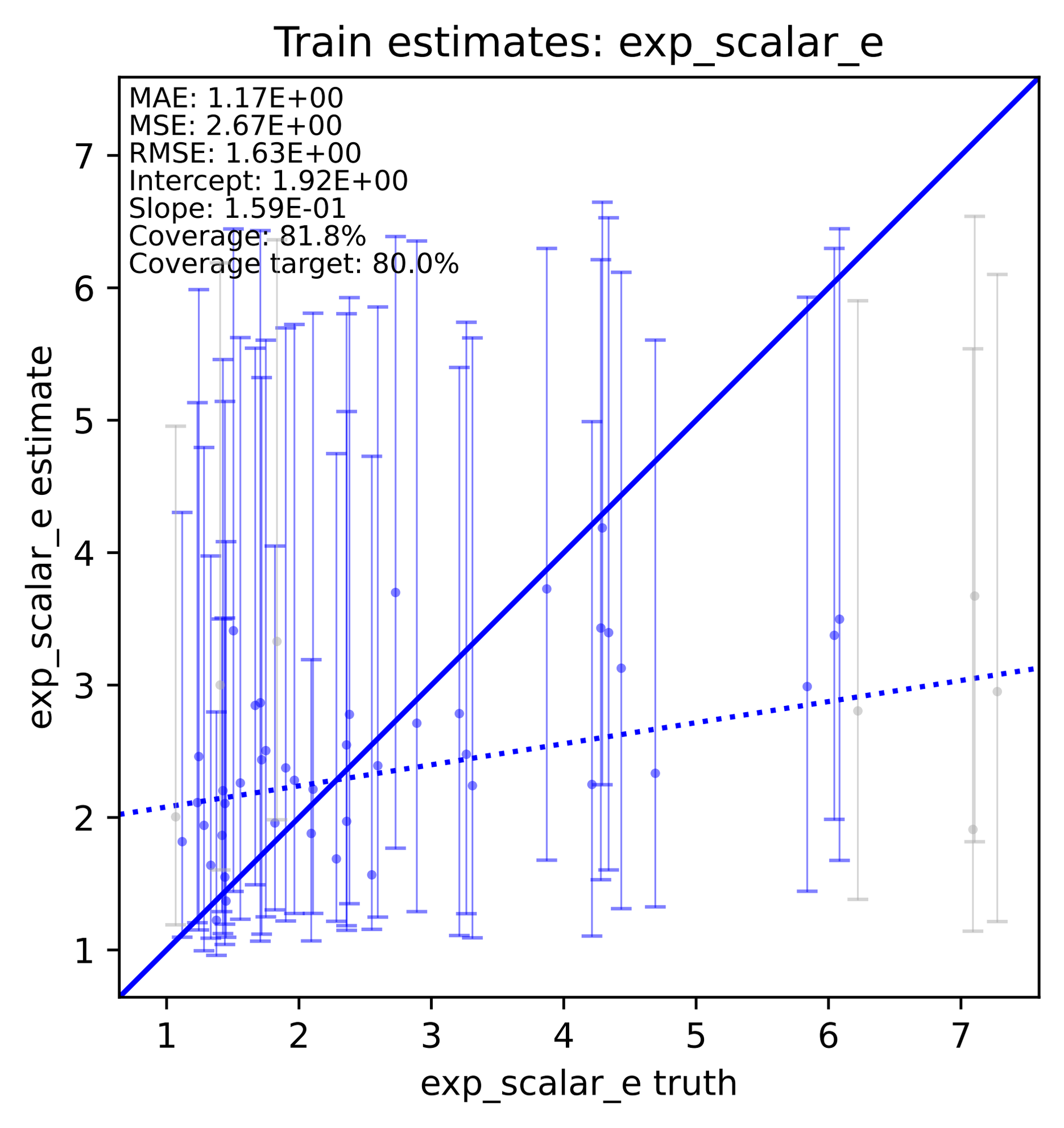

PCA: training labels
PC2 (17%)
% variance
PC
PC3 (16%)
PC4 (16%)
PC1 (37%)
PC2 (17%)
PC3 (16%)

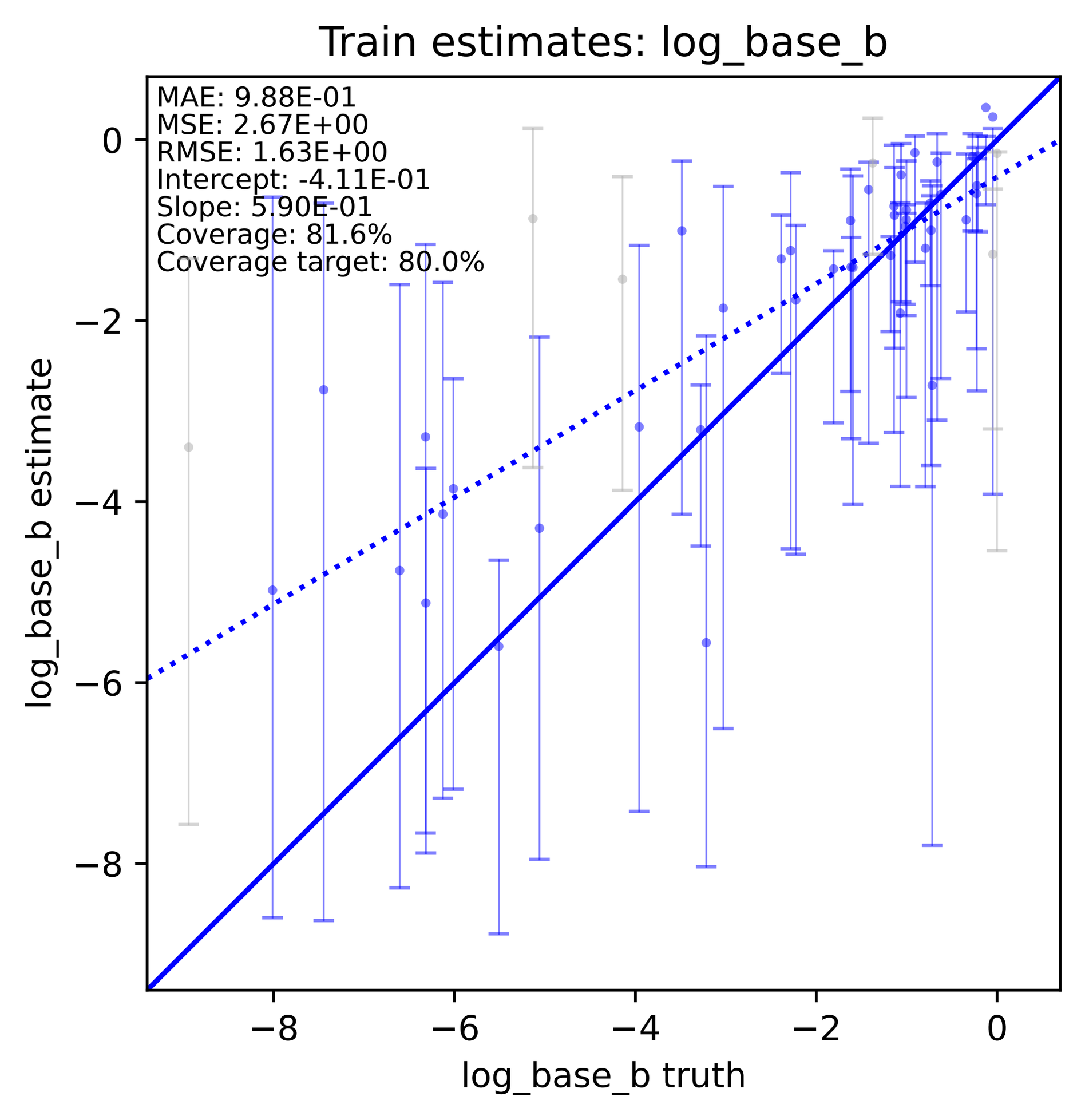

PCA: training phy. data
PC2 (9%)
% variance
PC
PC3 (4%)
PC4 (2%)
PC1 (27%)
PC2 (9%)
PC3 (4%)

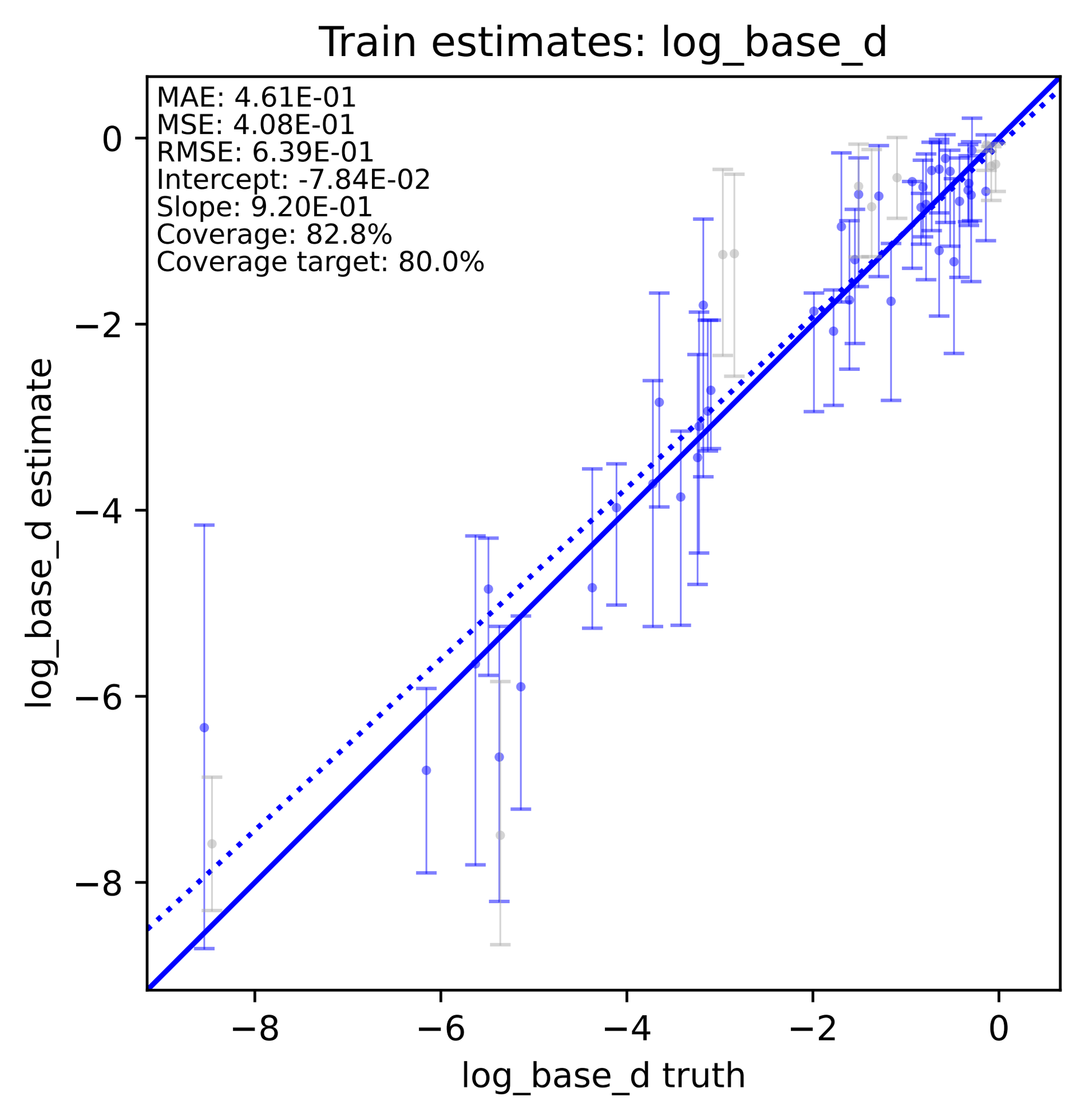

Density: training aux. data
N_bar
sim: 0.94 (0.59, 1.04)
emp: 0.99 (0.99, 0.99)
age_var
sim: 1.19 (-0.43, 2.76)
emp: 2.43 (2.43, 2.43)
brlen_var
sim: 0.72 (-0.99, 2.42)
emp: 1.97 (1.97, 1.97)
colless
sim: -1.25 (-1.60, -0.40)
emp: -1.22 (-1.22, -1.22)
f_dat_0
sim: 5.05 (1.58, 473.66)
emp: 4.00 (4.00, 4.00)
f_dat_1
sim: 2.24 (1.02, 118.96)
emp: 1.00 (1.00, 1.00)
log10_B1
sim: 1.84 (0.75, 2.19)
emp: 1.90 (1.90, 1.90)
log10_age_mean
sim: 0.50 (-0.19, 1.48)
emp: 1.36 (1.36, 1.36)
log10_brlen_mean
sim: 0.26 (-0.48, 1.23)
emp: 1.07 (1.07, 1.07)
log10_root_age
sim: 1.24 (0.46, 1.96)
emp: 1.81 (1.81, 1.81)
log10_tree_length
sim: 2.55 (1.48, 3.47)
emp: 3.56 (3.56, 3.56)
n_dat_0
sim: 6.89 (3.97, 10.04)
emp: 6.45 (6.45, 6.45)
n_dat_1
sim: 5.90 (3.33, 8.65)
emp: 5.06 (5.06, 5.06)
num_taxa
sim: 130.00 (11.00, 284.00)
emp: 158.00 (158.00, 158.00)
prop_taxa
sim: 1.00 (1.00, 1.00)
emp: 1.00 (1.00, 1.00)
treeness
sim: -0.28 (-0.40, -0.07)
emp: -0.44 (-0.44, -0.44)
Density
Variable

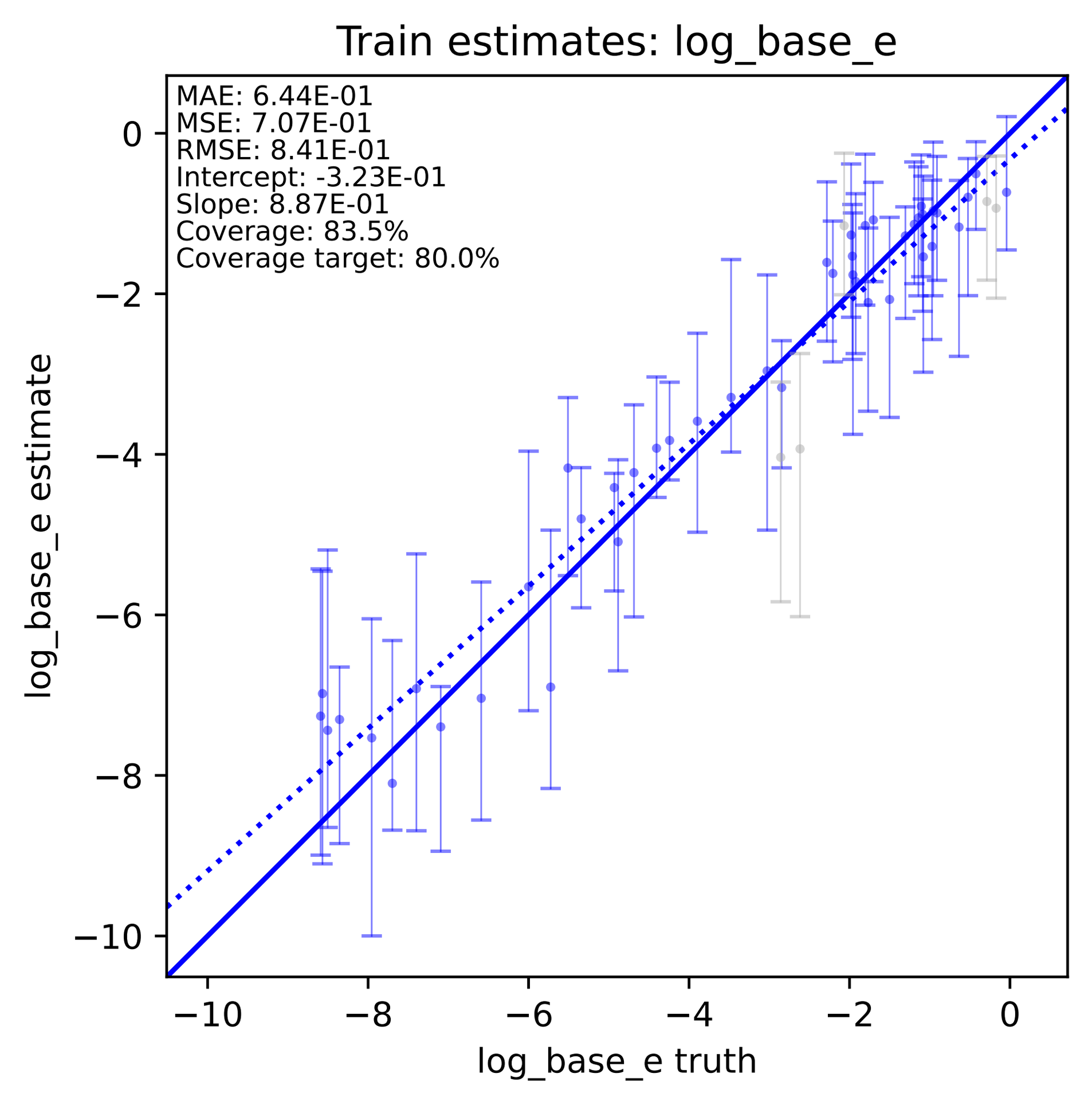

Density: training labels
exp_scalar_d_j
sim: 0.39 (0.14, 0.95)
emp: 0.16 (0.16, 0.16)
exp_scalar_e
sim: 2.33 (1.04, 6.89)
emp: 2.64 (2.64, 2.64)
log_base_b
sim: -1.52 (-8.65, -0.05)
emp: -1.07 (-1.07, -1.07)
log_base_d
sim: -1.17 (-8.43, -0.03)
emp: -6.25 (-6.25, -6.25)
log_base_e
sim: -2.92 (-8.87, -0.18)
emp: -6.77 (-6.77, -6.77)
log_base_w
sim: -1.08 (-7.89, -0.03)
emp: -3.34 (-3.34, -3.34)
Density
Variable

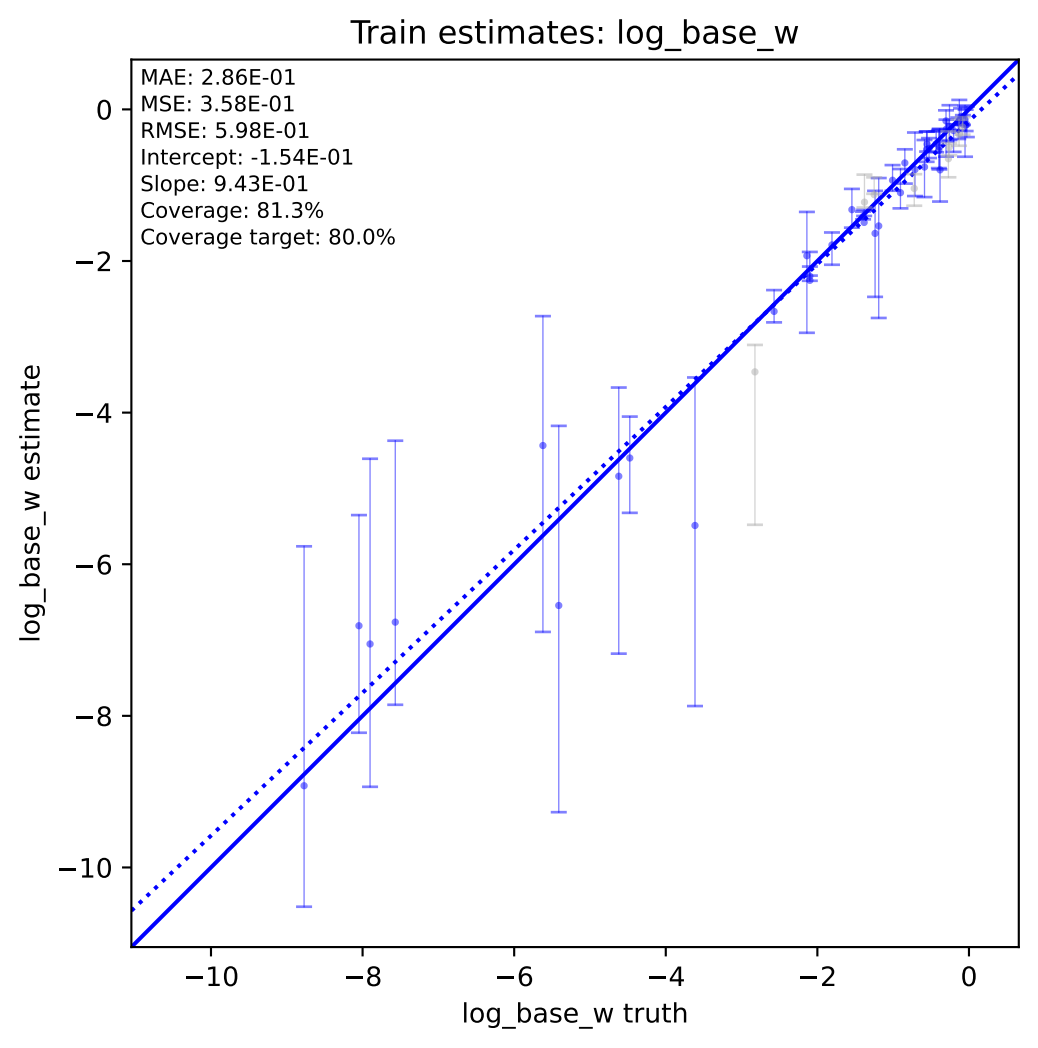

Train estimates: exp_scalar_d_j
MAE: 1.27E-01
MSE: 3.02E-02
RMSE: 1.74E-01
Intercept: 1.97E-01
Slope: 5.09E-01
Coverage: 85.1%
Coverage target: 80.0%
exp_scalar_d_j estimate
exp_scalar_d_j truth
0.2
0.4
0.6
0.8
1.0

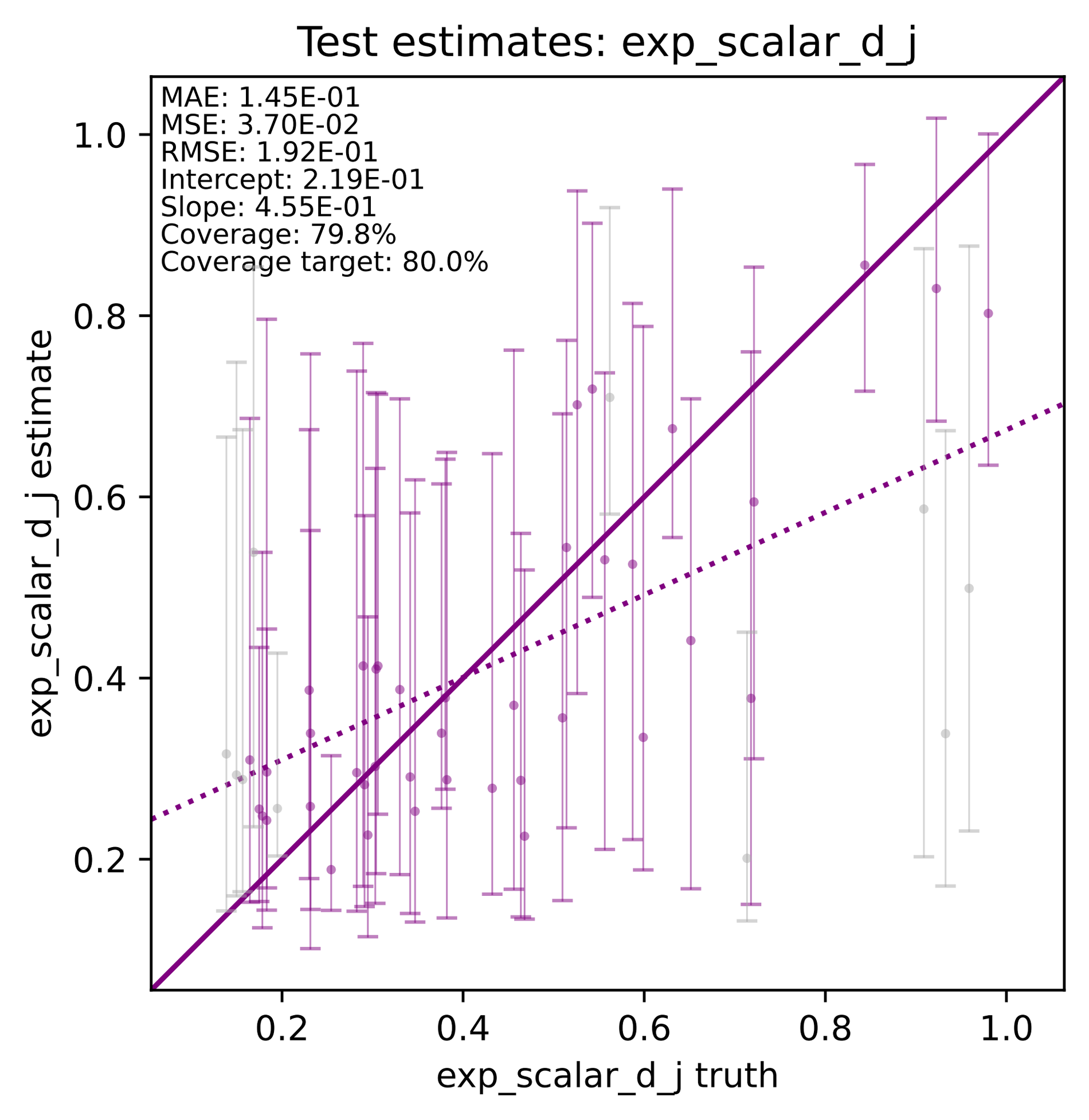

Train estimates: exp_scalar_e
MAE: 1.17E+00
MSE: 2.67E+00
RMSE: 1.63E+00
Intercept: 1.92E+00
Slope: 1.59E-01
Coverage: 81.8%
Coverage target: 80.0%
exp_scalar_e estimate
exp_scalar_e truth
1
2
3
4
5
6
7

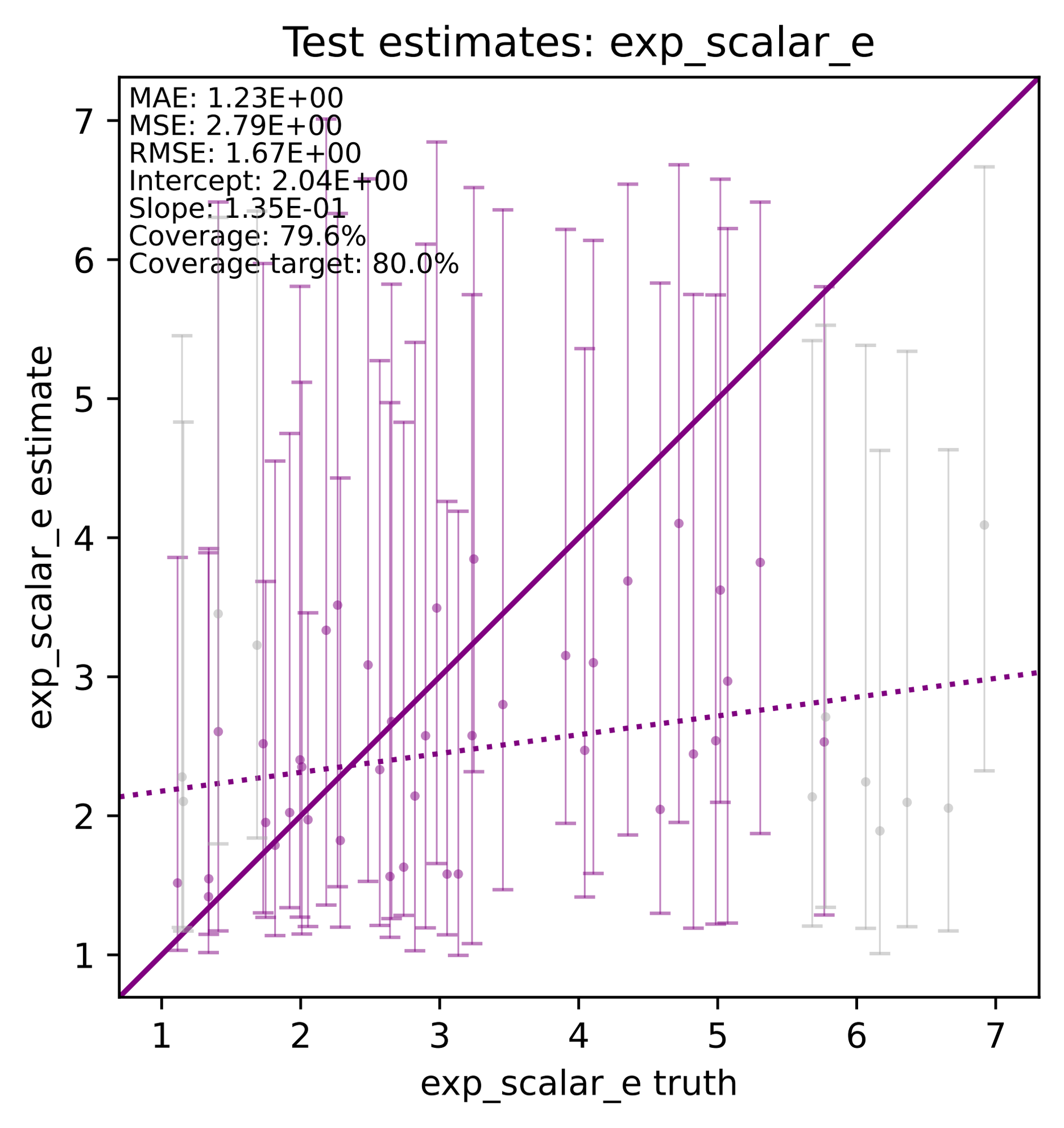

Train estimates: log_base_b
MAE: 9.88E-01
MSE: 2.67E+00
RMSE: 1.63E+00
Intercept: -4.11E-01
Slope: 5.90E-01
Coverage: 81.6%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

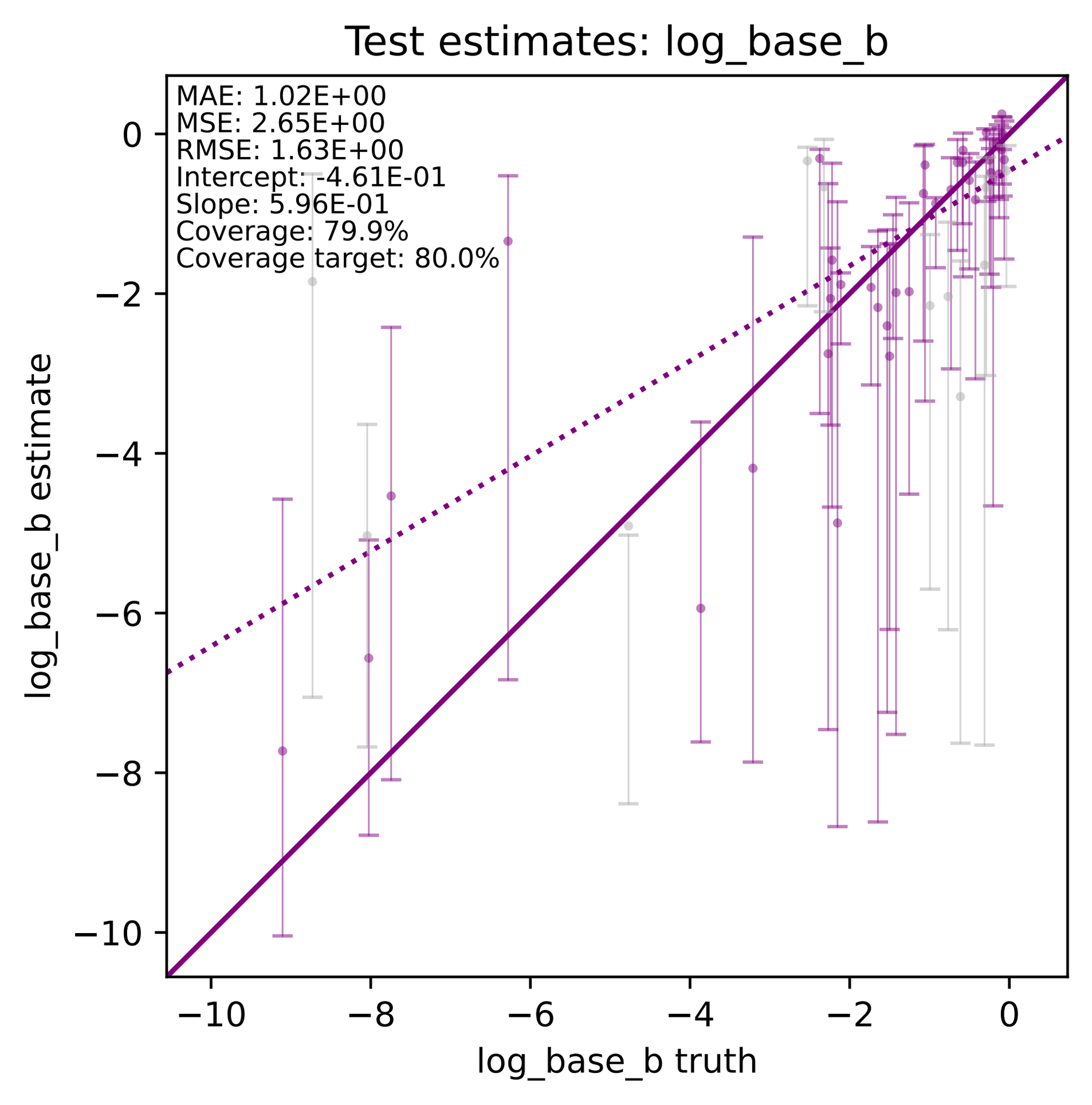

Train estimates: log_base_d
MAE: 4.61E-01
MSE: 4.08E-01
RMSE: 6.39E-01
Intercept: -7.84E-02
Slope: 9.20E-01
Coverage: 82.8%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

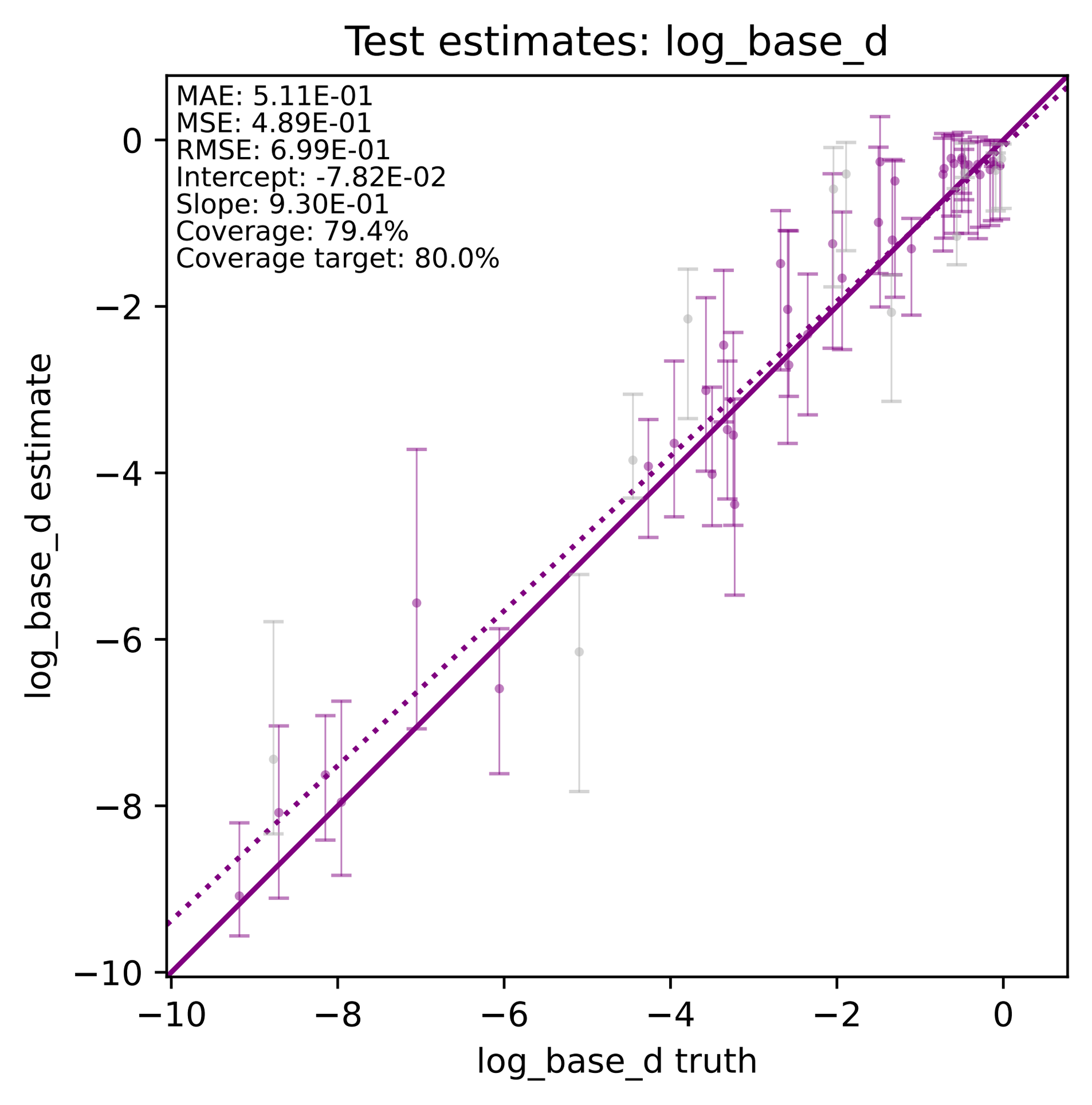

Train estimates: log_base_e
MAE: 6.44E-01
MSE: 7.07E-01
RMSE: 8.41E-01
Intercept: -3.23E-01
Slope: 8.87E-01
Coverage: 83.5%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

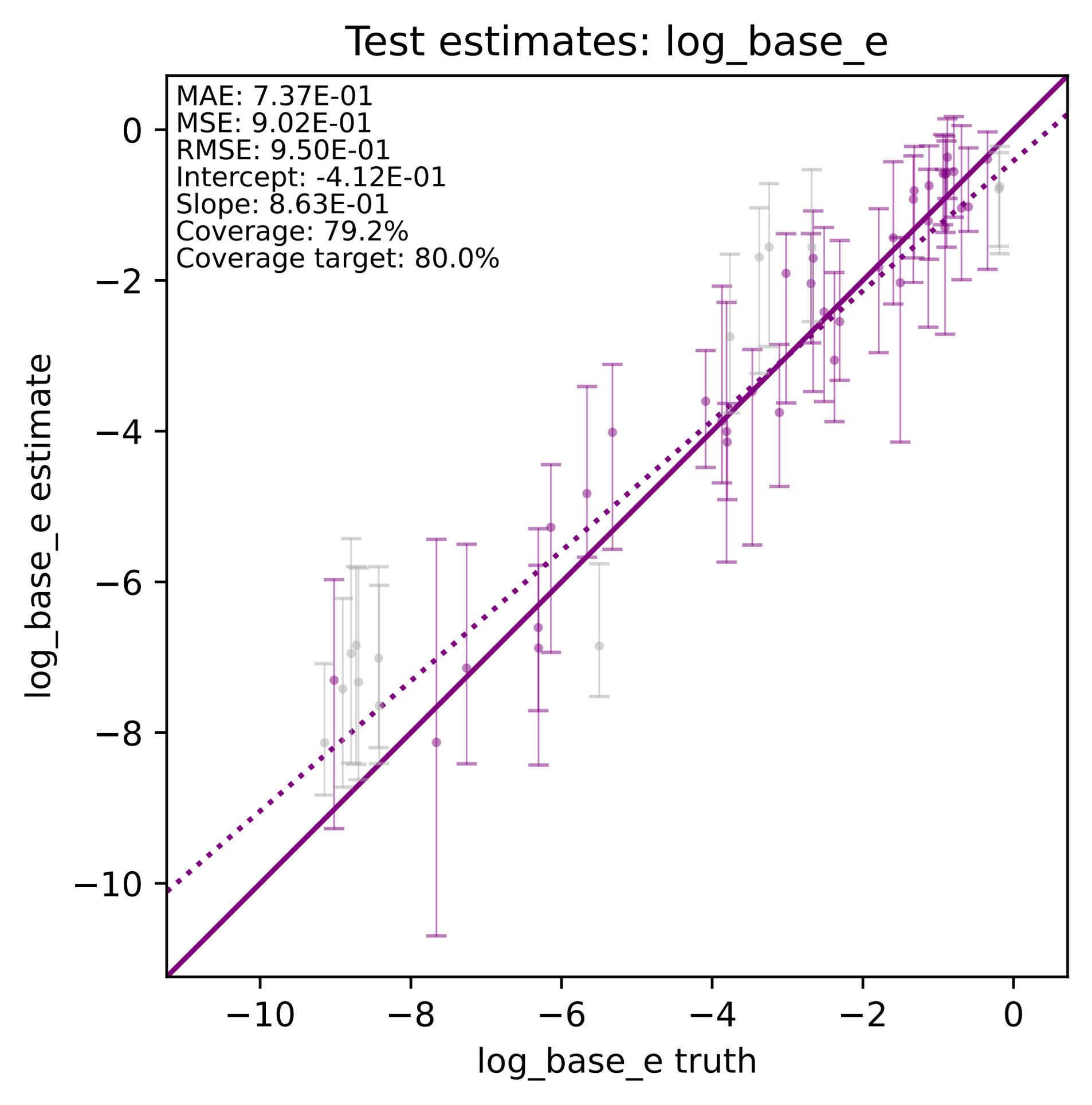

Train estimates: log_base_w
MAE: 2.86E-01
MSE: 3.58E-01
RMSE: 5.98E-01
Intercept: -1.54E-01
Slope: 9.43E-01
Coverage: 81.3%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

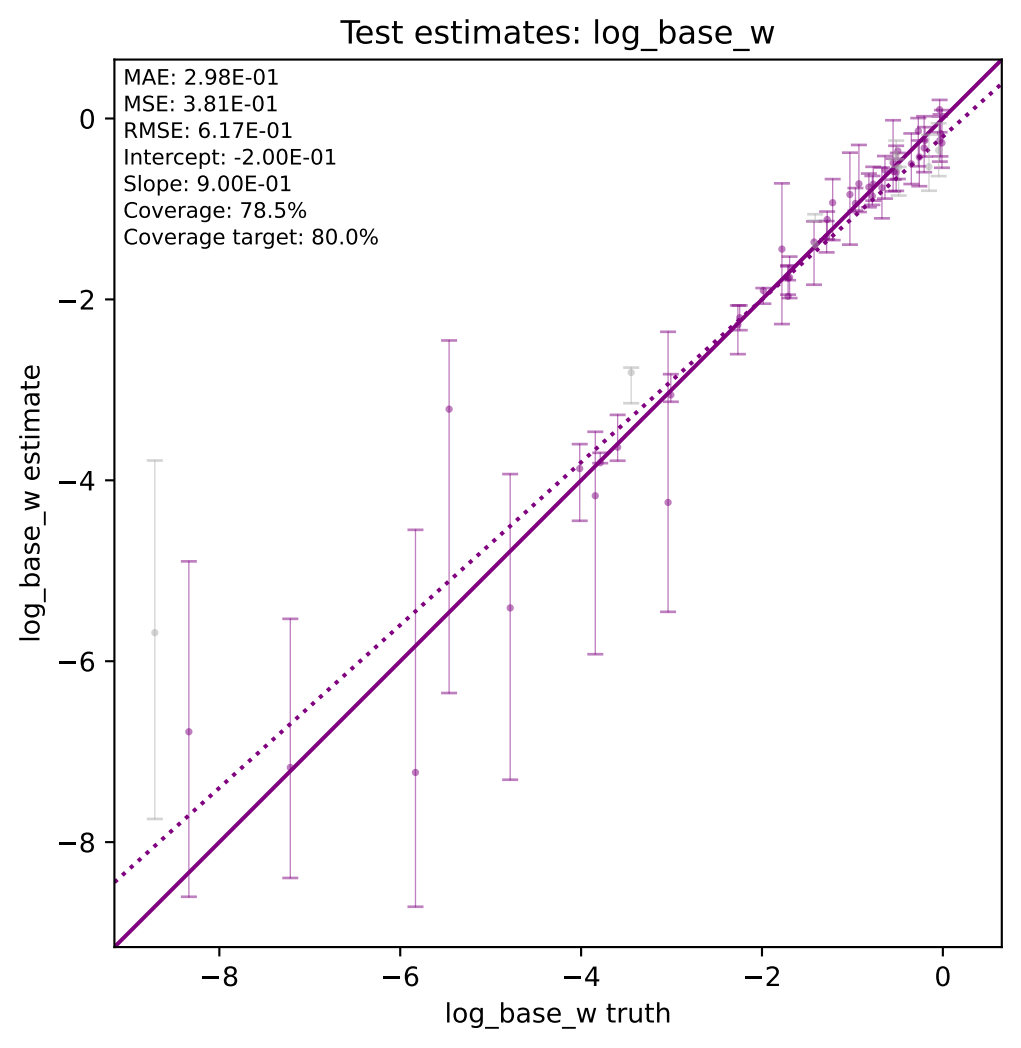

Test estimates: exp_scalar_d_j
MAE: 1.45E-01
MSE: 3.70E-02
RMSE: 1.92E-01
Intercept: 2.19E-01
Slope: 4.55E-01
Coverage: 79.8%
Coverage target: 80.0%
exp_scalar_d_j estimate
exp_scalar_d_j truth
0.2
0.4
0.6
0.8
1.0

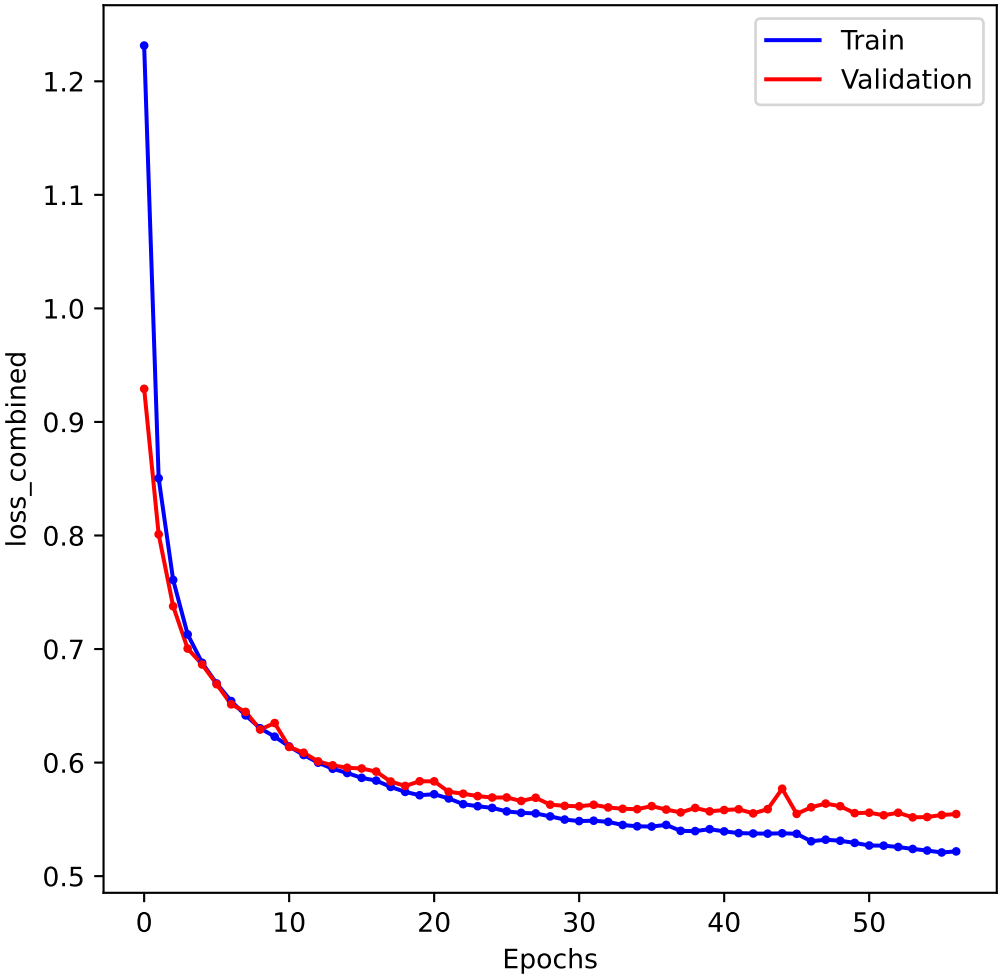

Test estimates: exp_scalar_e
MAE: 1.23E+00
MSE: 2.79E+00
RMSE: 1.67E+00
Intercept: 2.04E+00
Slope: 1.35E-01
Coverage: 79.6%
Coverage target: 80.0%
exp_scalar_e estimate
exp_scalar_e truth
1
2
3
4
5
6
7

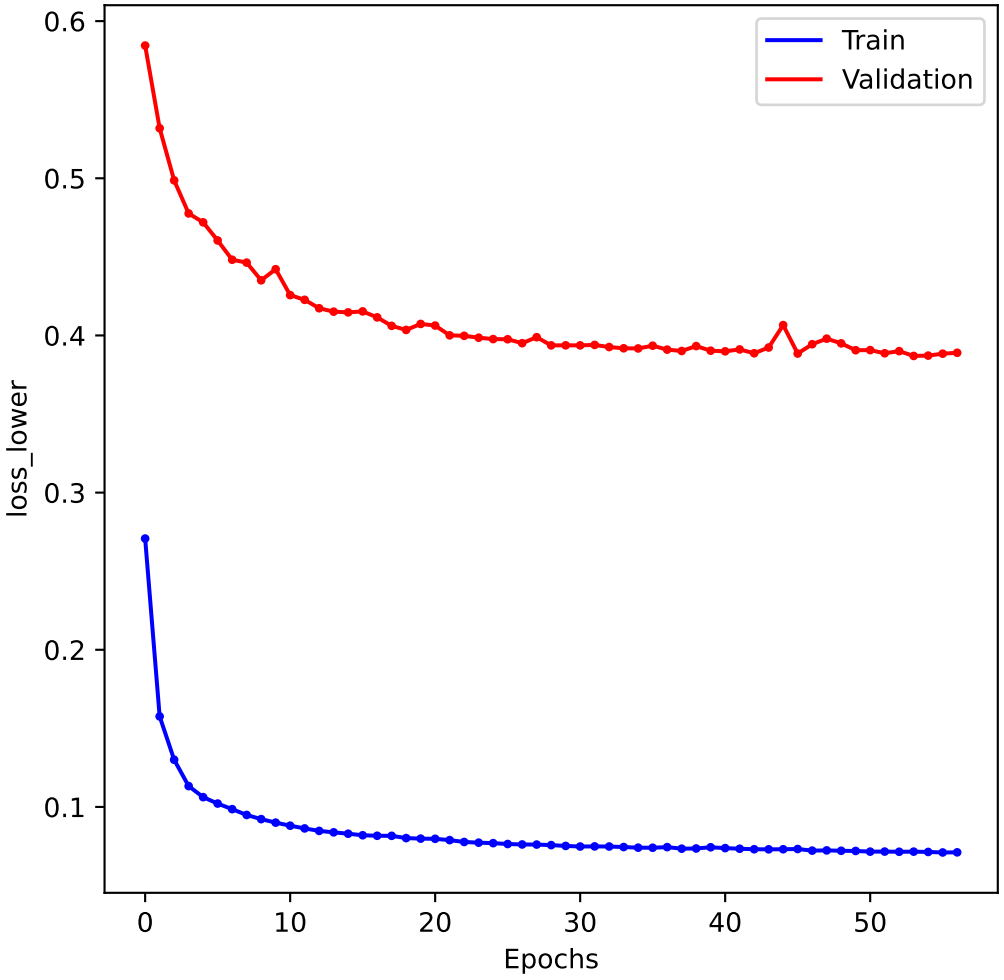

Test estimates: log_base_b
MAE: 1.02E+00
MSE: 2.65E+00
RMSE: 1.63E+00
Intercept: -4.61E-01
Slope: 5.96E-01
Coverage: 79.9%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

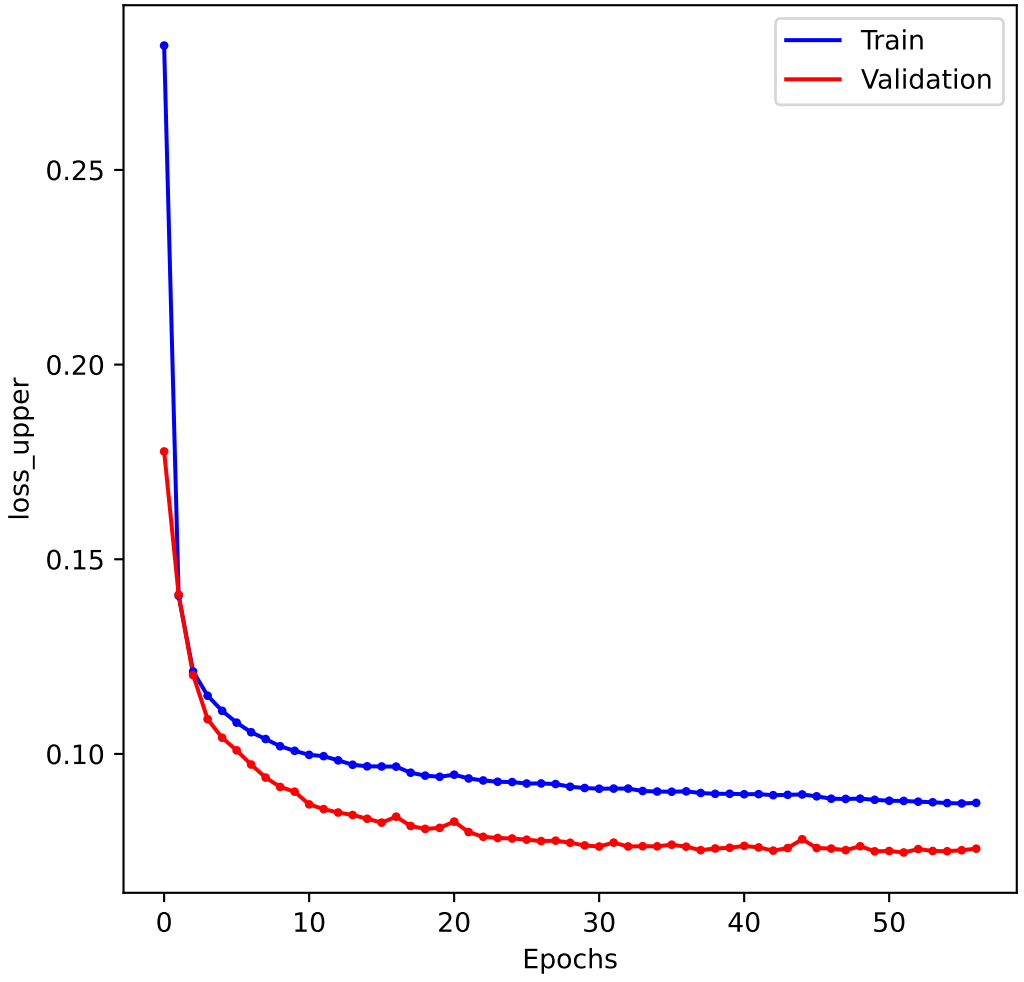

Test estimates: log_base_d
MAE: 5.11E-01
MSE: 4.89E-01
RMSE: 6.99E-01
Intercept: -7.82E-02
Slope: 9.30E-01
Coverage: 79.4%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

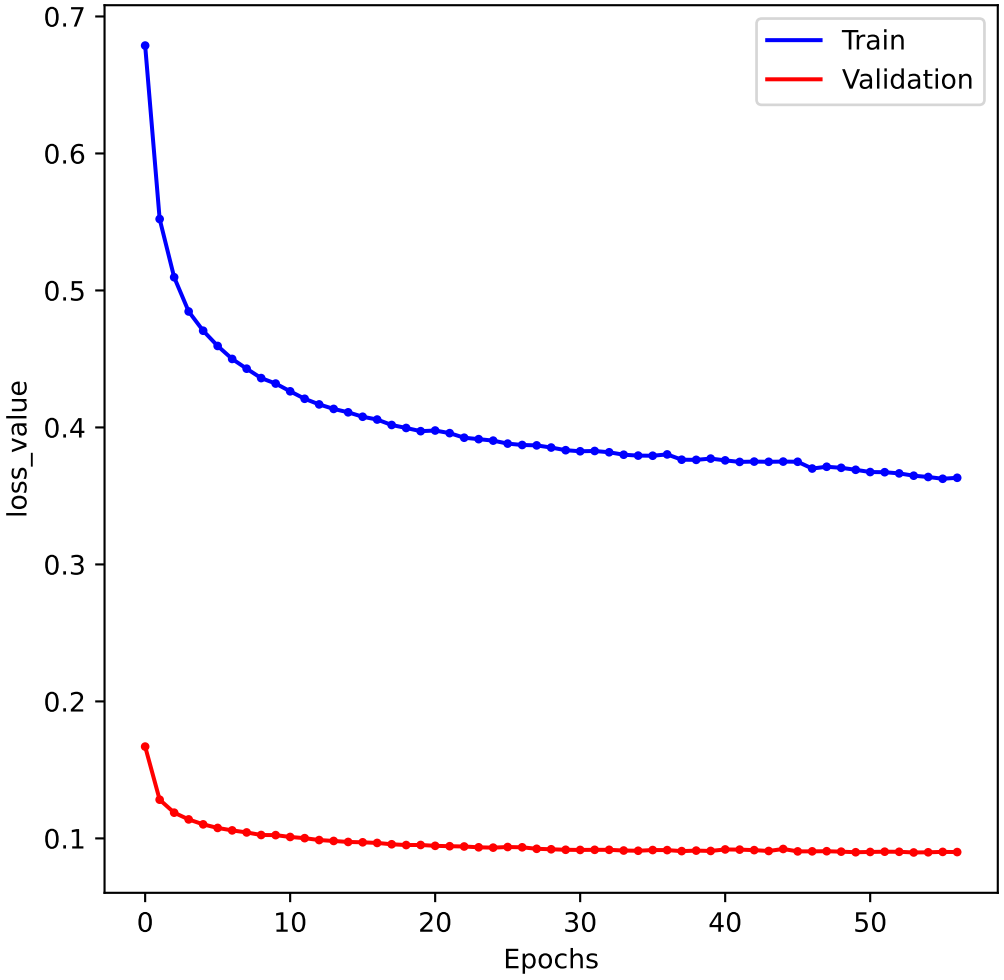

Test estimates: log_base_e
MAE: 7.37E-01
MSE: 9.02E-01
RMSE: 9.50E-01
Intercept: -4.12E-01
Slope: 8.63E-01
Coverage: 79.2%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

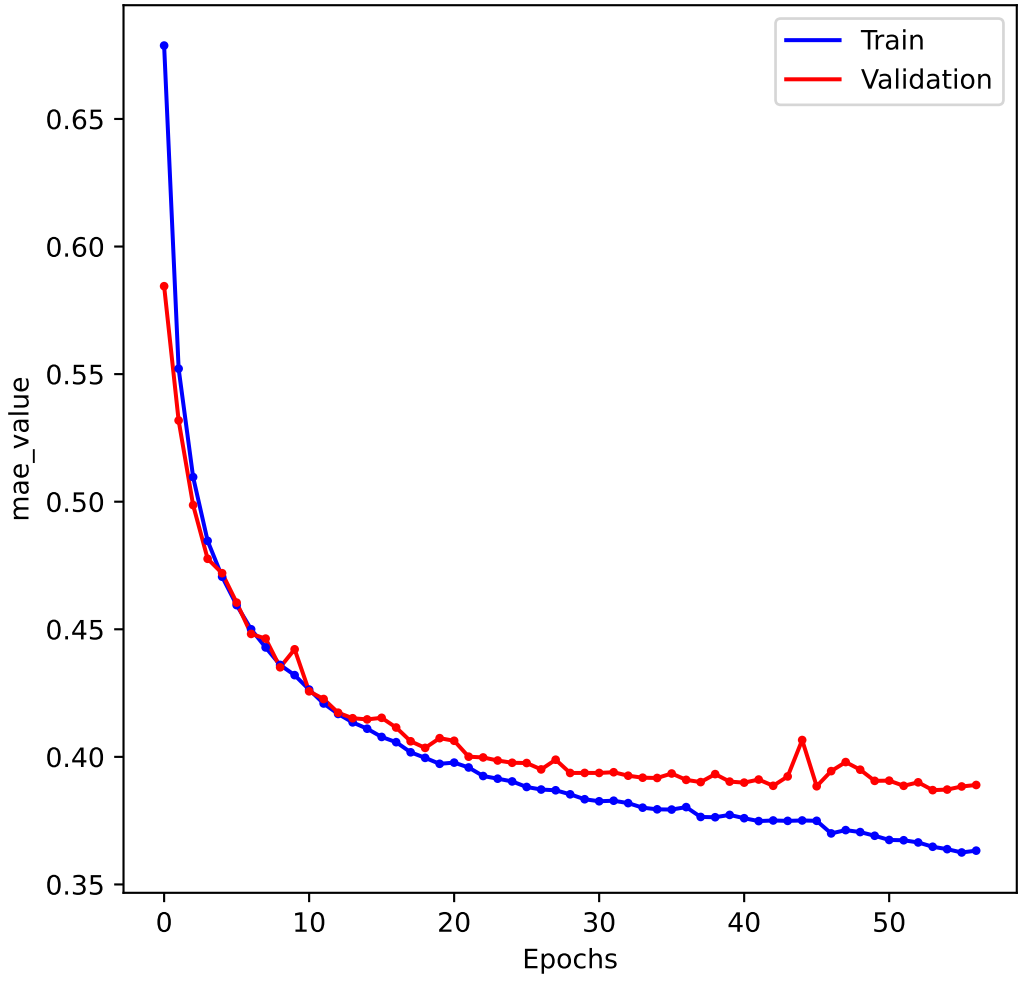

Test estimates: log_base_w
MAE: 2.98E-01
MSE: 3.81E-01
RMSE: 6.17E-01
Intercept: -2.00E-01
Slope: 9.00E-01
Coverage: 78.5%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

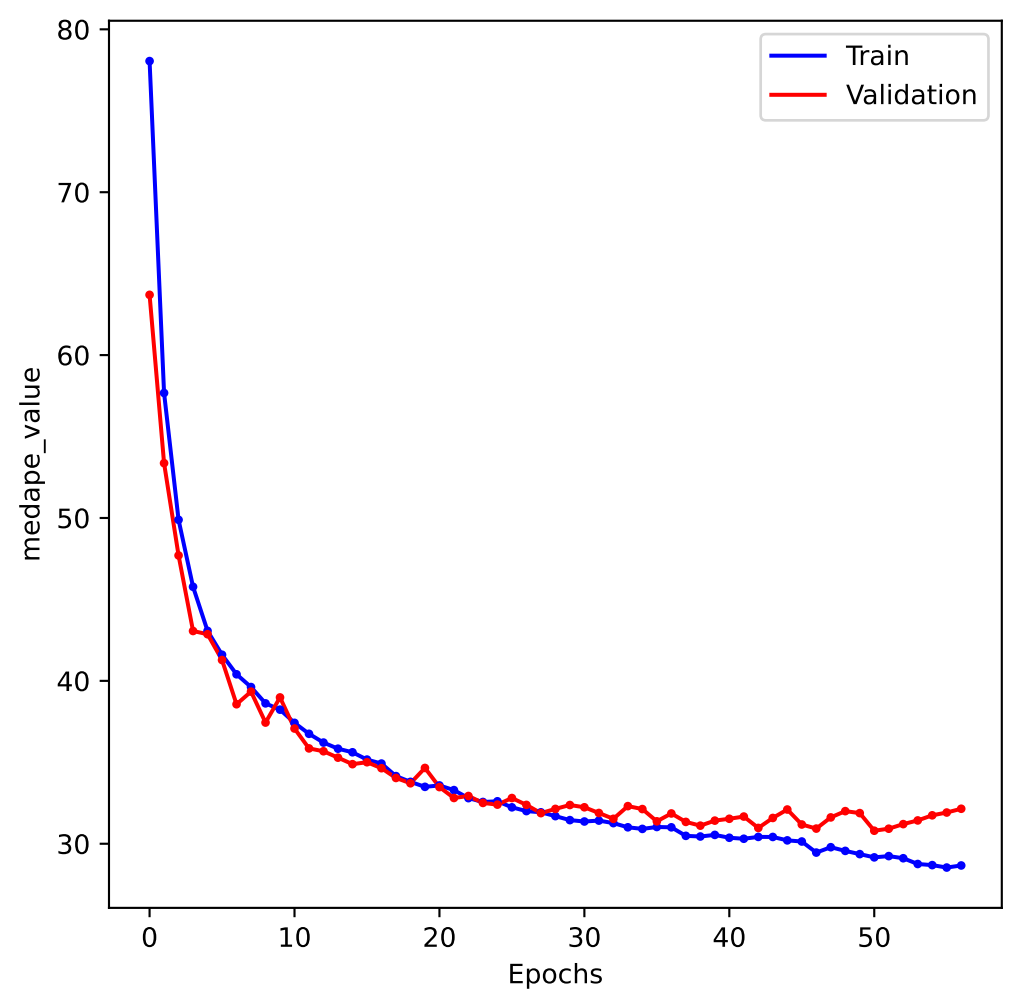

Train
Validation
loss_combined
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0
10
20
30
40
50
Epochs

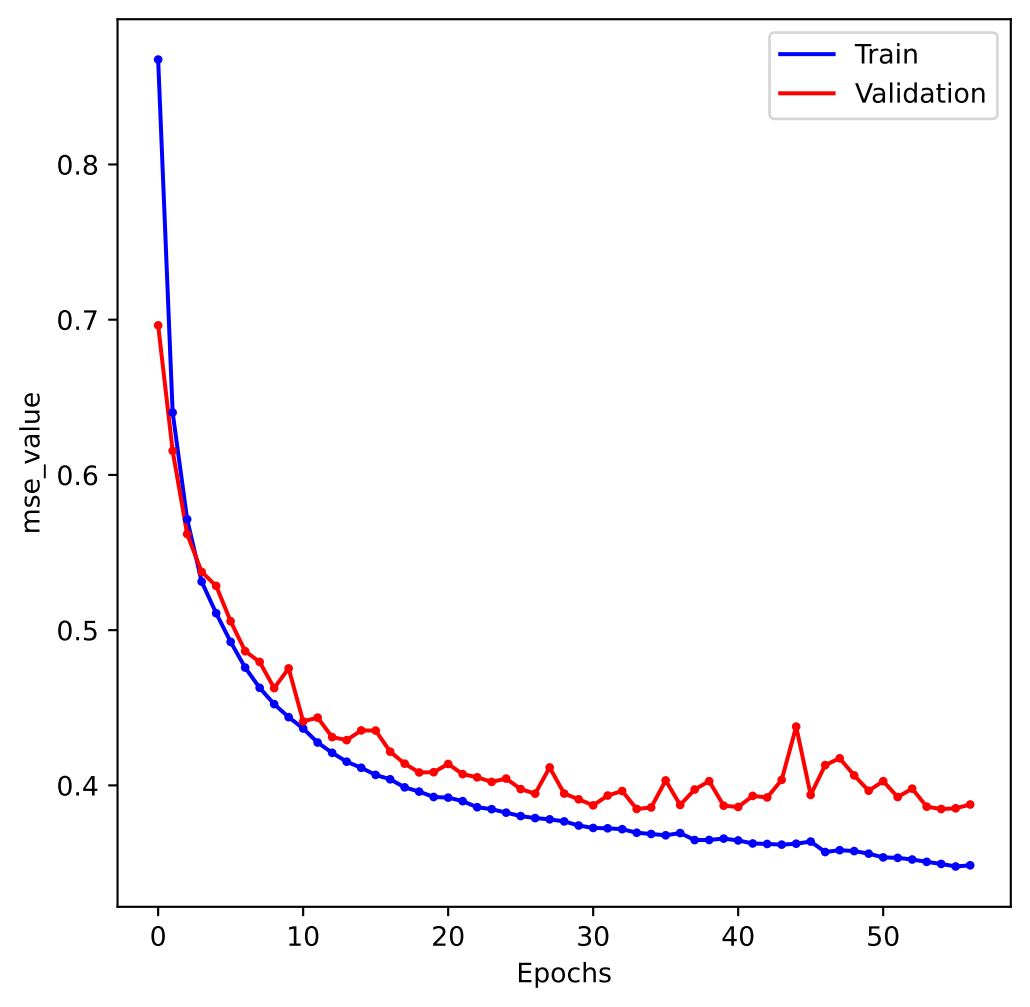

Train
Validation
loss_lower
0.6
0.5
0.4
0.3
0.2
0.1
0
10
20
30
40
50
Epochs

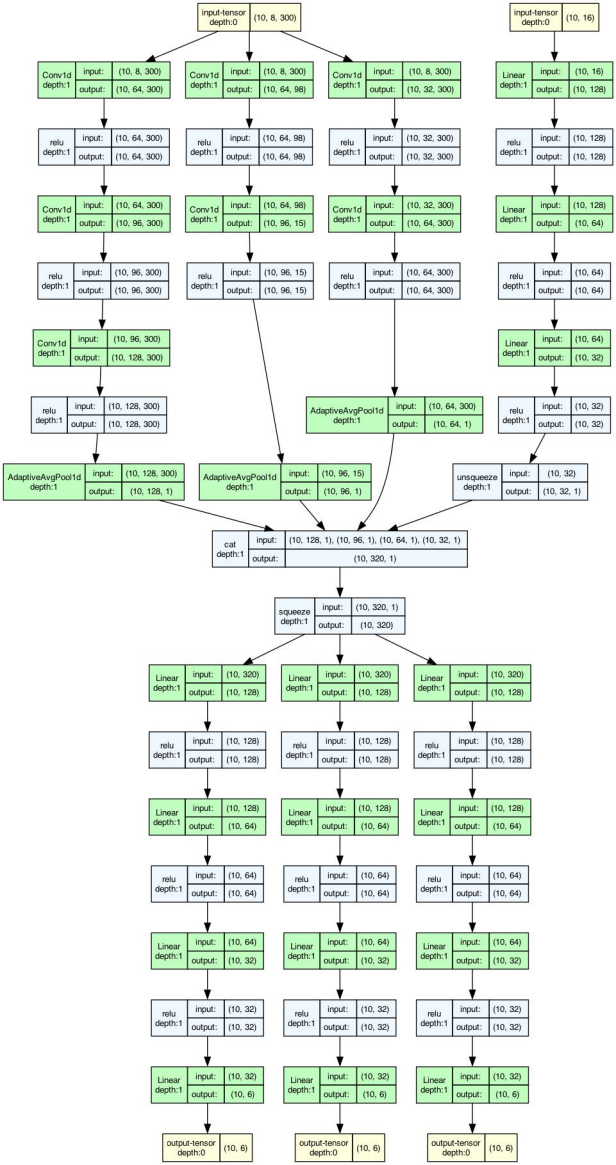

Train
Validation
0.25
0.20
0.15
0.10
loss_upper
0
10
20
30
40
50
Epochs


Train
Validation
loss_value
Epochs
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
10
20
30
40
50


Train
Validation
mae_value
0.65
0.60
0.55
0.50
0.45
0.40
0.35
0
10
20
30
40
50
Epochs


Train
Validation
medape_value
Epochs
80
70
60
50
40
30
0
10
20
30
40
50


Train
Validation
mse_value
0.8
0.7
0.6
0.5
0.4
0
10
20
30
40
50
Epochs


input-tensor depth:0
(10, 8, 300)
input-tensor depth:1
(10, 16)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 300)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 98)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 32, 300)
Linear depth:1
input: (10, 16)
output: (10, 128)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64, 98)
output: (10, 64, 98)
relu depth:1
input: (10, 32, 300)
output: (10, 32, 300)
relu depth:1
input: (10, 128)
output: (10, 128)
Conv1d depth:1
input: (10, 64, 300)
output: (10, 96, 300)
Conv1d depth:1
input: (10, 64, 98)
output: (10, 96, 15)
Conv1d depth:1
input: (10, 32, 300)
output: (10, 64, 300)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 96, 300)
output: (10, 96, 300)
relu depth:1
input: (10, 96, 15)
output: (10, 96, 15)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64)
output: (10, 64)
Conv1d depth:1
input: (10, 96, 300)
output: (10, 128, 300)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 128, 300)
output: (10, 128, 300)
AdaptiveAvgPool1d depth:1
input: (10, 64, 300)
output: (10, 64, 1)
relu depth:1
input: (10, 32)
output: (10, 32)
AdaptiveAvgPool1d depth:1
input: (10, 128, 300)
output: (10, 128, 1)
AdaptiveAvgPool1d depth:1
input: (10, 96, 15)
output: (10, 96, 1)
unsqueeze depth:1
input: (10, 32)
output: (10, 32, 1)
cat depth:1
input: (10, 128, 1), (10, 96, 1), (10, 64, 1), (10, 32, 1)
output: (10, 320, 1)
squeeze depth:1
input: (10, 320, 1)
output: (10, 320)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
Linear depth:1
input: (10, 32)
output: (10, 6)
Linear depth:1
input: (10, 32)
output: (10, 6)
Linear depth:1
input: (10, 32)
output: (10, 6)
output-tensor depth:0
(10, 6)
output-tensor depth:0
(10, 6)
output-tensor depth:0
(10, 6)